Differentiable Fit of Empirical Distributions by Bernstein Phase Types

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idea behind the paper

Abstract. The empirical cumulative distribution function (ECDF) is an efficient estimator of the distribution of sampled data. Yet, it has discontinuities at each step (where probability mass is concentrated) which result in a probability density function (PDF) with discrete spikes preventing its use in models that require continuous PDFs. Bernstein approximants bridge this gap, yielding smooth estimators for bounded ECDFs and their PDFs. In this paper, we consider a strategy of the literature to select the order of Bernstein polynomials (BP) over bounded support, and we extend the approach to Bernstein phase types (BPH) in order to approximate distributions with unbounded support. Preliminary experiments confirm the effectiveness of the heuristics in balancing the bias-variance tradeoff, giving insight into the relation between the sample size and the Bernstein order. Comparative experiments with Gaussian kernel density estimator show that the BPH approximant achieves better or comparable performance on a benchmark of Erlang PDFs with expected value equal to 1, while guaranteeing a low memory footprint thanks to its closed-form representation. We provide a replication package to support the reproducibility of our experimental results.

Keywords: Empirical Cumulative Distribution Function, Bernstein Polynomials, Bernstein Phase Types, Gaussian Kernel Density Estimation

1 Introduction

parametric vs non-parametric approach.

In most practical applications, the distributions of stochastic quantities are not known in advance, but they must be estimated from independent observations. Parametric approaches select a family of distributions and fit the distribution parameters from sample data, e.g., to maximize likelihood or to match statistic properties of the data; in contrast, nonparametric approaches construct an estimate of the distribution from the data itself, with tunable assumptions on the smoothness of the real distribution. For example, kernel density estimation (KDE) [2,14,8] is a popular nonparametric approach where the real probability density function (PDF) is estimated as the sum of kernel functions (e.g., Gaussian or Uniform PDFs with zero mean) centered at each data point; the *bandwidth* parameter (e.g., the variance of the kernel functions) is used to control smoothness.

general PH introduction

Among parametric approaches, the family of phase-type (PH) distributions [17,12] has received significant attention for its power and flexibility. PH distributions are defined as the time to absorption in Markov chains; they provide a flexible tradeoff between approximation accuracy and complexity, which is obtained by varying the number of transient states, referred to as *phases*, that can be visited before reaching the absorbing state. PH distributions have closed-form densities and Laplace transforms, enjoy closure under finite mixtures and convolutions, and are dense in the set of positive-valued distributions (i.e., they can approximate any positive-valued distribution with arbitrary accuracy, provided enough phases). When used in stochastic models, PH distributions result in Markov chains with regular structures amenable to efficient analysis through matrix analytic methods [18,15]. Fitting methods have been proposed to find PH parameters (transition rates and initial state probabilities) that maximize likelihood [1,4] or that match moments [5,22], tail behavior [7,20], or both [9].

More recently, an approach was proposed to approximate PDFs [13] and cumulative distribution functions (CDFs) [10,11] using *Bernstein exponentials* (BEs), i.e., linear combinations of Bernstein polynomials (BP) [19,21] where the support $x \in [0,1]$ is mapped to $y \in [0,\infty)$ through the change of variable $x = e^{-y}$. This approach results in a subclass of acyclic PH distributions called Bernstein phase type (BPH) distributions, which preserve shape properties of approximated CDFs (including monotonicity, upper and lower bounds, and exact limit values at 0 and $+\infty$), while allowing fast, closed-form fitting of CDFs using only their values at specific points (specifically, one for each phase). BPH

When applied to the approximation of the empirical cumulative distribution function (ECDF) derived from sample data, the selection of the number of phases of the BPH distribution plays an important role in the bias-variance tradeoff [6]: using more phases reduces the approximation error (i.e., the error due to the choice of model) but increases the estimation error (i.e., the error due to the limited size of the data sample). Since a BE of the order N results in a BPH with N phases, selecting the number of phases is equivalent to the selection of the order of a Bernstein exponential. Prior work analyzed the properties of Bernstein polynomials for the approximation of ECDFs; in particular, if the true CDF has a continuous PDF f that is Lipschitz of order 1, the approximation of the ECDF with BPs results in a PDF that converges uniformly to f, provided that the order N of the BP is such that $N = o(M/\log(M))$, where $N \to \infty$ and $M \to \infty$ is the number of samples that define the ECDF [3]. Higher order expansions for the asymptotic mean-squared error are provided in [16].

In this paper, we apply the analysis of BPs of [3] for the order selection of BEs, i.e., to select the number of phases of BPH estimators of an ECDF. Our experiments show that, if the sample size M is sufficiently representative of the true CDF, then the accuracy of the estimated PDF increases with N only until $N \approx M/\log(M)$; conversely, if the sample size M is insufficient, increasing the paper intention

Bernstein order N does not improve estimation accuracy. We also compare our parametric approach based on BPHs with nonparametric KDE using Gaussian kernels, showing that BPH estimation achieves better or similar accuracy, while maintaining a low memory footprint thanks to its compact representation, which requires a number of parameters equal to the Bernstein order N (instead of the dataset size M). The paper is accompanied by a replication package available at https://github.com/oris-tool/bph-fitting (with open source license AGPLv3).

The rest of the paper is organized as follows. First, we illustrate BP and BPH approximants, and we summarize the approach of [3] to select the order of BPs approximating PDFs with bounded support (Section 2). Then, we define our experimental setup using BPHs to approximate PDFs with unbounded support (Section 3) and we present experimental results (Section 4). Finally, we draw our conclusions and discuss next steps (Section 5).

2 Bernstein Approximants

2.1 Bernstein Polynomials

Given a CDF $F\colon [0,1] \to [0,1]$ and the ECDF

approximation of CDF and PDF

$$F_M(x) = \frac{1}{M} \sum_{i=1}^{M} I\{X_i \leq x\} \tag{1}$$

obtained from a random sample $X_1, X_2, \ldots, X_M$, the BP approximant of F_M with order N is defined as

$$B_N(F_M; x) := \sum_{n=0}^{N} F_M\left(\frac{n}{N}\right) \binom{N}{n} x^n (1-x)^{N-n} \,. \tag{2}$$

Note that $B_N(F_M; x)$ preserves the values of F_M at 0 and 1, the lower and upper bounds (i.e., $l \leq F_M(x) \leq u \;\forall\, x \in [0,1] \implies l \leq B_N(F_M; x) \leq u \;\forall\, x \in [0,1]$), and monotonicity (i.e., $B_N(F_M; x)$ is monotonically nondecreasing given that $F_M(x)$ is monotonically nondecreasing). According to this, $B_N(F_M; x)$ is a valid CDF for all $N > 0$. Moreover, the approximation error $|F_M(x) - B_N(F_M; x)|$ converges uniformly to 0 over the support $[0,1]$ as $N \to \infty$ (i.e., $\forall \epsilon > 0, \exists \bar{N} \in \mathbb{N}$ such that $N > \bar{N} \implies |F_M(x) - B_N(F_M; x)| < \epsilon \;\forall\, x \in [0,1]$). To approximate the PDF f of F, the derivative of $B_N(F_M; x)$ for $x \in [0,1]$ can be calculated as

$$B'_N(F_M; x) = \sum_{n=1}^{N} \left(F_M\left(\frac{n}{N}\right) - F_M\left(\frac{n-1}{N}\right) \right) n \binom{N}{n} x^{n-1} (1-x)^{N-n} \,. \tag{3}$$

Proof. By linearity of differentiation, $B'_N(F_M;x) = \sum_{n=0}^{N} F_M\left(\frac{n}{N}\right) \cdot T'_{N,n}(x)$ where $T_{N,n}(x) = \binom{N}{n} x^n (1-x)^{N-n}$ and, for $n = 0, 1, \dots, N$ and $x > 0$,

$$\begin{aligned}
T'_{N,n}(x) &= n\binom{N}{n} x^{n-1}(1-x)^{N-n} - (N-n)\binom{N}{n} x^n (1-x)^{N-(n+1)} \\
&= \frac{n}{x}\binom{N}{n} x^n (1-x)^{N-n} - \frac{n+1}{x}\binom{N}{n+1} x^{n+1}(1-x)^{N-(n+1)} \\
&= \Big(n\, T_{N,n}(x) - (n+1)\, T_{N,n+1}(x)\Big)/x\,.
\end{aligned}$$

Note that $T_{N,N+1}(x) = 0$ and that we used the identity

$$(N-n)\binom{N}{n} = \frac{N!}{n!(N-(n+1))!} = (n+1)\binom{N}{n+1}\,.$$

Then, we have that:

$$\begin{aligned}
B'_N(F_M;x) &= \sum_{n=0}^{N} F_M\left(\frac{n}{N}\right) \cdot \Big(n\, T_{N,n}(x) - (n+1)\, T_{N,n+1}(x)\Big)/x \\
&= \left(\sum_{n=0}^{N} F_M\left(\frac{n}{N}\right) n\, T_{N,n}(x)/x\right) - \left(\sum_{n=0}^{N} F_M\left(\frac{n}{N}\right)(n+1)\, T_{N,n+1}(x)/x\right) \\
&= \left(\sum_{n=1}^{N} F_M\left(\frac{n}{N}\right) n\, T_{N,n}(x)/x\right) - \left(\sum_{n=1}^{N} F_M\left(\frac{n-1}{N}\right) n\, T_{N,n}(x)/x\right) \\
&= \sum_{n=1}^{N} \left(F_M\left(\frac{n}{N}\right) - F_M\left(\frac{n-1}{N}\right)\right) n\, T_{N,n}(x)/x \\
&= \sum_{n=1}^{N} \left(F_M\left(\frac{n}{N}\right) - F_M\left(\frac{n-1}{N}\right)\right) n \binom{N}{n} x^{n-1}(1-x)^{N-n}.
\end{aligned}$$

Note that the result is also valid at $x = 0$ (where $B'_N(F_M;0) = N[F_M(1/N) - F_M(0)]$) and, by continuity, for all $x \in [0,1]$. □

If F has a compact support $[a,b]$, i.e., $F\colon [a,b] \to [0,1]$, its BP approximant of order N is obtained from Eq. (2) by the change of variables $x = (y-a)/(b-a)$:

CDF and PDF approzimation if F:[a,b]\to[0,1]

$$\begin{aligned}
B_N^{[a,b]}(F_M;y) &= B_N(F_M;(y-a)/(b-a)) \\
&= \sum_{n=0}^{N} F_M\left(a + \frac{n}{N}(b-a)\right)\binom{N}{n}\frac{(y-a)^n(b-y)^{N-n}}{(b-a)^N}, \qquad (4)
\end{aligned}$$

where F_M is the ECDF obtained from a random sample of F of size M. Since the change of variables is continuous and strictly monotonic, $B_N^{[a,b]}(F_M;y)$ inherits the shape preservation properties of $B_N(F_M;x)$. Moreover, to approximate the PDF f of F, the derivative of $B_N^{[a,b]}(F_M;y)$ for $y \in [a,b]$ can be calculated as

$$\frac{d}{dy}B_N^{[a,b]}(F_M;y) = \frac{d}{dy}B_N(F_M;(y-a)/(b-a)) = \frac{1}{b-a}B'_N(F_M;(y-a)/(b-a))\,.$$

2.2 Bernstein Phase Types

If F has unbounded support $[0,\infty)$, i.e., $F\colon [0,\infty) \to [0,1]$, the BPH approximant of order N of its ECDF is obtained from Eq. (2) through the change of variable $x = e^{-y}$: CDF and PDF approzimation if F:[0,\infrty]\to[0,1]

$$BPH_N(F_M; y) = \sum_{n=0}^{N} F_M\left(\log\left(\frac{N}{n}\right)\right)\binom{N}{n} e^{-ny}(1-e^{-y})^{N-n}, \tag{5}$$

where F_M is the ECDF obtained from a random sample of F of size M. The division by zero for $n = 0$ is resolved by considering the limiting value of $F_M(x)$ as x tends to infinity, i.e., $F_M\left(\log \frac{N}{0}\right) := \lim_{x\to\infty} F_M(x) = 1$; for $n = N$, we have $F_M(\log(N/N)) = F_M(0)$ which is 0 if the ECDF has no samples equal to 0. Since the change of variables is continuous and strictly monotonic, $BPH_N(F_M; y)$ inherits the shape preservation properties of $B_N(F_M; x)$. Moreover, the derivative of $BPH_N(F_M; y)$, which we use to approximate the PDF f of F, is equal to

$$\begin{aligned} BPH'_N(F_M; y) &= \frac{d}{dy} B_N(F_M; e^{-y}) \\ &= -e^{-y} B'_N(F_M; e^{-y}) \\ &= \sum_{n=1}^{N} \left(F_M\left(\log\left(\frac{N}{n-1}\right)\right) - F_M\left(\log\left(\frac{N}{n}\right)\right)\right) \\ &\qquad \cdot n\binom{N}{n} e^{-ny}(1-e^{-y})^{N-n}. \end{aligned} \tag{6}$$

2.3 Fitting Approach

Related work [3] studies the asymptotic properties observed when the true CDF $F(x)$ and PDF $f(x)$ with support $[0,1]$ are approximated using BPs $B_N(F_M; x)$, where N is the order of the BP and $F_M(x)$ is an instance of the ECDF of F using M samples. Specifically, if F is continuous over $[0,1]$, then $B_N(F_M; x)$ converges w.p.1 uniformly to F as $N, M \to \infty$. More interestingly, if F is continuous and differentiable, and f is Lipschitz of order 1, then $B_N(F_M; x)$ converges w.p.1 uniformly to F as $N \to \infty$, provided that $M^{2/3} \le N \le (M/\log(M))^2$. A similar result is shown for $B'_N(F_M; x)$, proving that, if the CDF F has a continuous PDF f that is Lipschitz of order 1, then $B'_N(F_M; x)$ converges w.p.1 uniformly to f as $N, M \to \infty$, provided that $N = o(M/\log(M))$.

We extend the order selection approach proposed in [3] to the BPH estimators of a PDF f with unbounded support $[0,\infty)$. In particular, we experimentally show that, if the sample size M is sufficiently representative of the CDF F, then the accuracy of the BPH estimator of the PDF f increases as the Bernstein order N increases, until the threshold $M/\log(M)$ is reached, beyond which the approximation accuracy no longer improves and indeed becomes worse. Conversely, if the sample size M is insufficient, then increasing the Bernstein order N does not improve the accuracy of the BPH estimator of f, which actually remains poor.

3 Experimental Setup

We consider Erlang CDFs $\mathcal{E}_K(x)$ having both shape and rate parameters equal to $K \in \{2, 4, 8, 16\}$, thus resulting in expected value equal to 1 and variance $1/K$. As described in Section 2.2, we approximate the PDF of $\mathcal{E}_K(x)$ by the derivative of the BPH approximant of its ECDF, i.e., $bph_N(\mathcal{E}_{K,M}; x) := BPH'_N(\mathcal{E}_{K,M}; x)$, where $\mathcal{E}_{K,M}(x)$ is the ECDF of $\mathcal{E}_K(x)$ for a random sample of size M.

For each value of K, we compute $bph_N(\mathcal{E}_{K,M}; x)$ for each Bernstein order $N \in \{8, 16, 32, 64, 128, 256\}$ and for each number of samples $M \in \{27, 68, 163, 381, 866, 1938\}$. The values of N correspond to the bound $M/\log(M)$ defined in [3], i.e., $8 \approx 27/\log(27)$, $16 \approx 68/\log(68)$, $\ldots$, and $256 \approx 1938/\log(1938)$.

We evaluate the accuracy of $bph_N(\mathcal{E}_{K,M}; x)$ with respect to the PDF of $\mathcal{E}_K(x)$, denoted by $\mathcal{E}'_K(x)$, through a discrete approximation of the Kullback-Leibler Divergence (KLD):

$$D_{KL}\left(\mathcal{E}'_K(x) \,||\, bph_N(\mathcal{E}_{K,M}; x)\right) = \sum_{x \in \mathcal{X}} \mathcal{E}'_K(x) \cdot \log\left(\frac{\mathcal{E}'_K(x)}{bph_N(\mathcal{E}_{K,M}; x)}\right) \tag{7}$$

where $\mathcal{X}$ is the set of 500 equidistant time points covering the interval $[10^{-9}, l]$ such that $1 - \mathcal{E}_K(l) = 10^{-3}$ (i.e., the probability that the Erlang exceeds l is lower than 10^{-3}). The evaluation of $bph_N(\mathcal{E}_{K,M}; x)$ and of its KLD is repeated for 100 random samples of size M (each resulting in a different $\mathcal{E}_{K,M}$); we report the mean value and the standard deviation of the KLD.

For each value of K and M, we compare our BPH approximant with Gaussian Kernel Density Estimation (KDE), which is defined as

$$kde_h(\mathcal{E}'_{K,M}; x) = \frac{1}{M} \sum_{m=1}^{M} \frac{1}{h\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x - x_m}{h}\right)^2\right) \tag{8}$$

where h is a smoothing parameter called *bandwidth*, selected according to Scott's rule $h = M^{-1/5}$. Also for KDE, we evaluate the KLD with respect to the ground truth PDF $\mathcal{E}'_K(x)$, i.e., $D_{KL}\left(\mathcal{E}'_K(x) \,||\, kde_h(\mathcal{E}'_{K,M}; x)\right)$.

4 Experimental Results

4.1 Results for Optimal Bernstein Order

Table 1 reports the accuracy of the BPH and KDE approximants of the considered Erlang PDFs with shape and rate equal to K, showing the mean value and the standard deviation of the KLD computed over 100 random samples of size M. Specifically, Table 1 considers the cases where the Bernstein order N is selected according to $N \approx M/\log(M)$ (as suggested by the results in [3]). For each value of K, the mean value and the standard deviation of the KLD decrease as the sample size M increases, given that increasing the number of samples provides a more representative statistical baseline of the ground truth

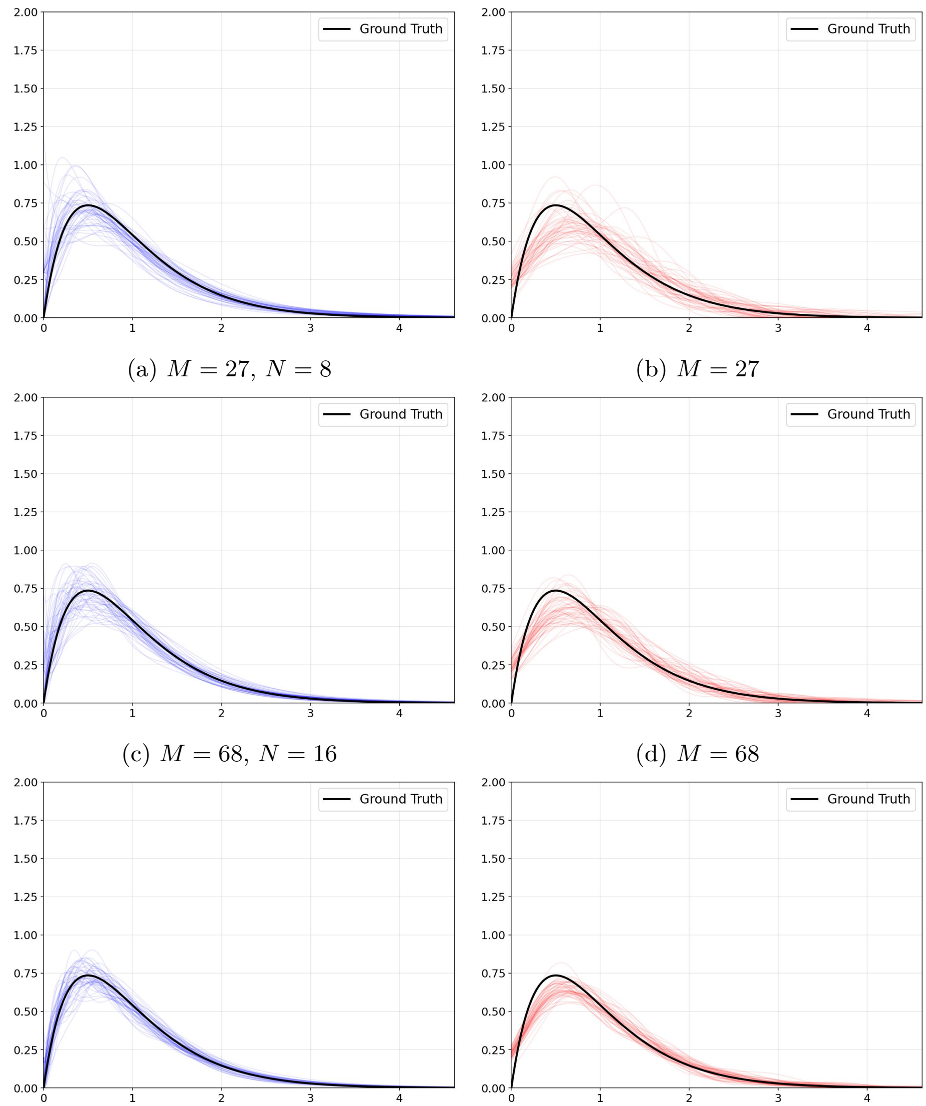
PDF. In particular, this trend is also evident in the plots of Figs. 1 and 2, which show that for $K = 2$ both BPHs and KDEs converge to the ground truth PDF as the sample size M grows. The same trend is also evident for $K \in \{4, 8, 16\}$; Figs. 3 and 4 show plots for $K = 4$ and $K = 16$, respectively.

As K increases, the considered Erlang PDFs become less right-skewed, and their probability mass concentrates over a narrower support, making the approximation more difficult for BPHs, also due to the fact that the points used to construct the approximant PDF are skewed towards the left end of the support. In fact, for small values of M, the mean KLD achieved by BPHs nearly doubles as K doubles, e.g., for $M = 27$, it is equal to 0.0381, 0.0609, 0.1431, and 0.2771 for $K = 2$, $K = 4$, $K = 8$, and $K = 16$, respectively. Similarly, the minimum

K	**M**	**N**	**KLD *bph* vs ground truth**		**KLD *kde* vs ground truth**	
			mean	standard deviation	mean	standard deviation
2	27	8	0.0381	0.0233	0.1181	0.1104
	68	16	0.0202	0.0131	0.0531	0.0315
	163	32	0.0118	0.0064	0.0305	0.0192
	381	64	0.0076	0.0035	0.0177	0.0071
	866	128	0.0051	0.0022	0.0103	0.0027
	1938	256	0.0031	0.0010	0.0068	0.0011
4	27	8	0.0609	0.0286	0.0953	0.0830
	68	16	0.0294	0.0181	0.0506	0.0282
	163	32	0.0141	0.0080	0.0291	0.0122
	381	64	0.0068	0.0039	0.0159	0.0062
	866	128	0.0040	0.0016	0.0098	0.0026
	1938	256	0.0029	0.0012	0.0062	0.0014
8	27	8	0.1431	0.0348	0.0855	0.0694
	68	16	0.0670	0.0173	0.0419	0.0209
	163	32	0.0255	0.0099	0.0224	0.0104
	381	64	0.0106	0.0052	0.0124	0.0055
	866	128	0.0045	0.0019	0.0066	0.0026
	1938	256	0.0024	0.0011	0.0037	0.0015
16	27	8	0.2771	0.0264	0.0713	0.0444
	68	16	0.1471	0.0177	0.0348	0.0265
	163	32	0.0697	0.0113	0.0183	0.0103
	381	64	0.0258	0.0069	0.0097	0.0041
	866	128	0.0089	0.0034	0.0052	0.0023
	1938	256	0.0035	0.0014	0.0029	0.0012

Table 1: Accuracy of BPHs (with order N) and KDEs in approximating Erlang PDFs with shape and rate equal to K: mean value and standard deviation of the KLD with respect to the ground truth, over 100 random samples of size M.

sample size M needed to achieve mean KLD in the order of $5 \cdot 10^{-3}$ increases with K, i.e., it is equal to 64, 64, 128, and 256 for $K = 2$, $K = 4$, $K = 8$, and $K = 16$, respectively. Nevertheless, such accuracy can be achieved for all the values of K considered in this experimental evaluation.



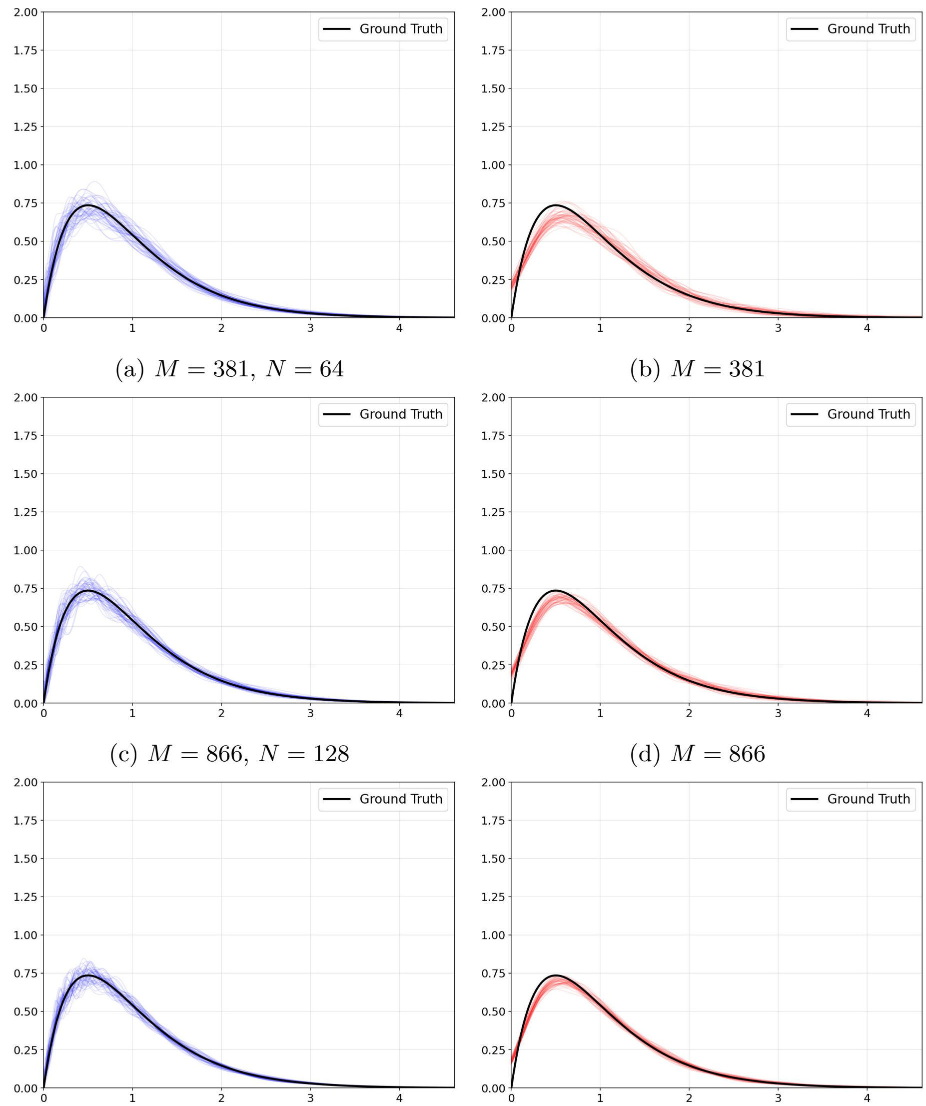
(a) $M = 27$, $N = 8$ (b) $M = 27$

(c) $M = 68$, $N = 16$ (d) $M = 68$

(e) $M = 163$, $N = 32$ (f) $M = 163$

Fig. 1: BPH approximants with order N (a,c,e) and KDE approximants (b,d,f) for 100 samples of size M of an Erlang PDF with shape and rate $K = 2$

Conversely, the approximation complexity does not increase with K for KDEs: in fact, for a given sample size M, KDEs exhibit a slight reduction in the KLD as K increases, due to the fact that the shape of the Erlang PDF more closely resembles a Gaussian PDF. Notably, when compared to KDE, the BPH approximant achieves lower mean value and standard deviation of the KLD for



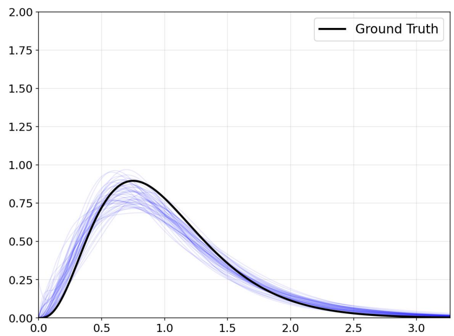
(a) $M = 381,\ N = 64$ (b) $M = 381$

(c) $M = 866,\ N = 128$ (d) $M = 866$

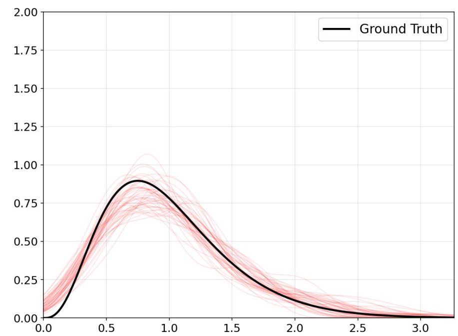
(e) $M = 1938,\ N = 256$ (f) $M = 1938$

Fig. 2: BPH approximants with order N (a,c,e) and KDE approximants (b,d,f) for 100 samples of size M of an Erlang PDF with shape and rate $K = 2$

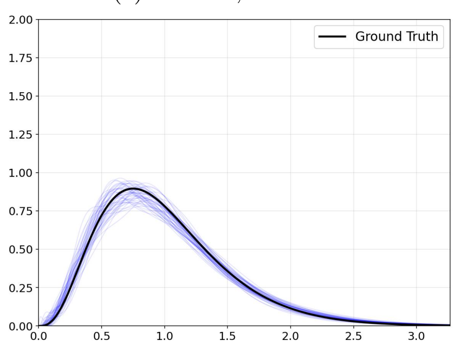
$K \in \{2, 4\}$ and all tested values of M. For $K = 8$, BPHs achieve better accuracy than KDEs for sample size $M \geq 381$. For $K = 16$, BPHs achieve slightly larger KLD with respect to KDEs, though still in the same order of magnitude for sample size $M \geq 866$, and in the order of $5 \cdot 10^{-2}$ for $M \in \{163, 381\}$. Overall,



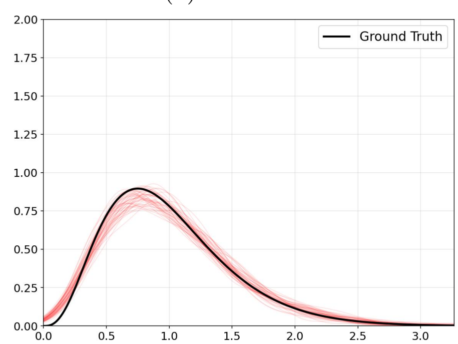
(a) $M = 68$, $N = 16$.



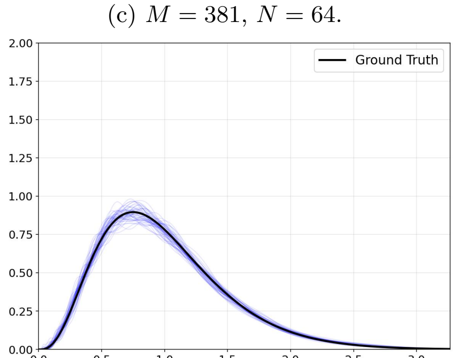
(b) $M = 68$.



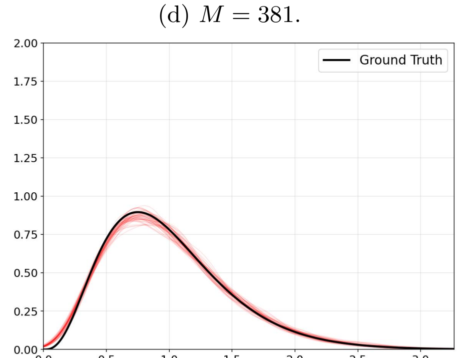
(c) $M = 381$, $N = 64$.



(d) $M = 381$.



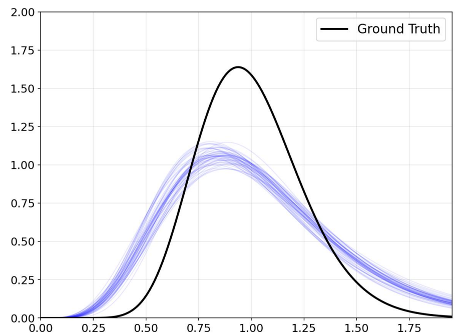
(e) $M = 1938$, $N = 256$.



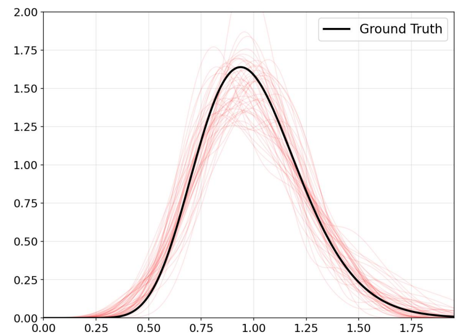
(f) $M = 1938$.

Fig. 3: BPH approximants with order N (a,c,e) and KDE approximants (b,d,f) for 100 samples of size M of an Erlang PDF with shape and rate $K = 4$

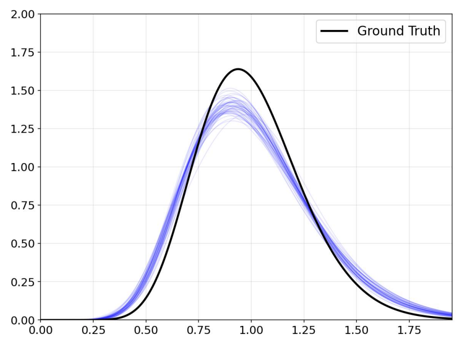
these results show the effectiveness of BPHs in approximating the considered Erlang PDFs.



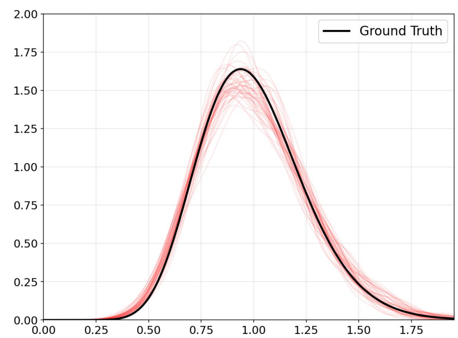
(a) $M = 68$, $N = 16$.



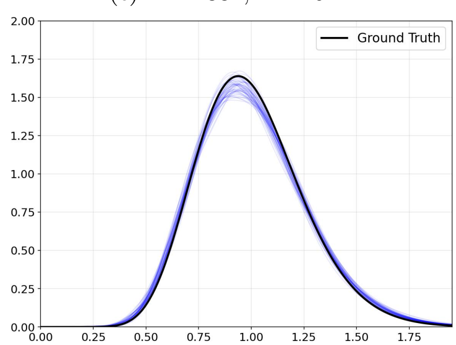
(b) $M = 68$.



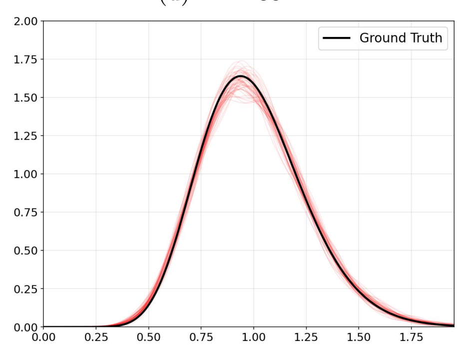
(c) $M = 381$, $N = 64$.



(d) $M = 381$.



(e) $M = 1938$, $N = 256$.

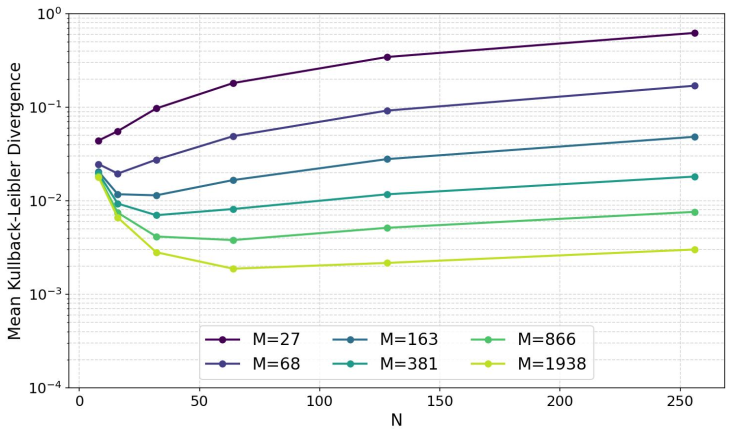


(f) $M = 1938$.

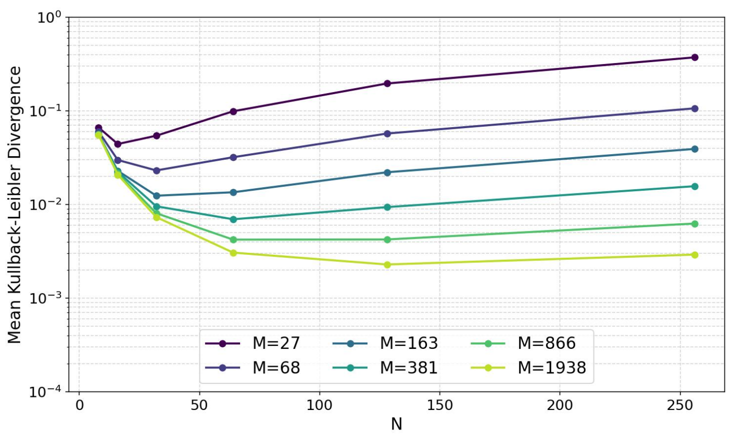
Fig. 4: BPH approximants with order N (a,c,e) and KDE approximants (b,d,f) for 100 samples of size M of an Erlang PDF with shape and rate $K = 16$

4.2 Results for Non-Optimal Bernstein Order

Figs. 5 and 6 illustrate the behavior of the BPH approximant for $K \in \{2, 4, 8, 16\}$ and for all combinations of sample size M and Bernstein order N. Specifically, in the case of the Erlang PDF with shape and rate $K = 2$ shown in Fig. 5a, the mean value of the KLD increases when N exceeds the threshold $M/\log(M)$, pointing out the effectiveness, for BPH approximants, of the strategy proposed in [3] to select the optimal order of BP approximants. For instance, for $M = 27$, corresponding to $N = 27/\log(27) \approx 8$, the mean KLD increases with N for any $N \geq 8$; for $M = 68$, corresponding to $N = 68/\log(68) \approx 16$, the mean KLD



(a) $K = 2$.

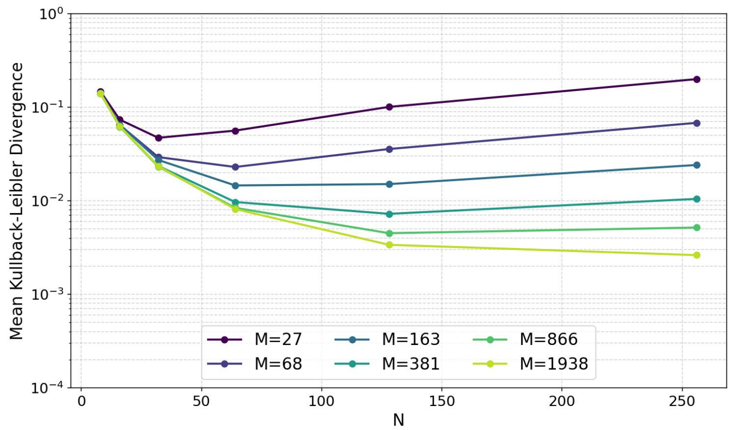


(b) $K = 4$.

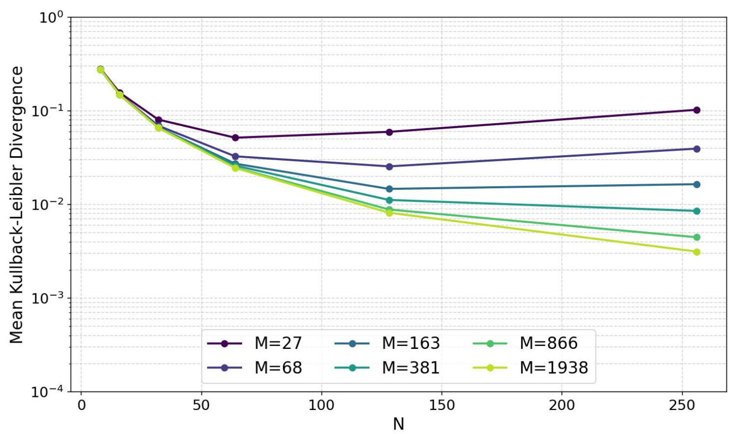
Fig. 5: Accuracy of BPHs (with order N) in approximating Erlang PDFs with shape and rate equal to K: mean value of the KLD with respect to the ground truth, over 100 random samples of size M.

slightly decreases as N increases from 8 to 16, and then increases with N for any $N \geq 16$; and so on. Also note that, when the sample size M is insufficient to provide a representative statistical baseline of the true PDF, then increasing N does not yield a good accuracy (i.e., the mean KLD is much greater than $5 \cdot 10^{-3}$) and indeed it worsens accuracy for $N > M/\log(M)$, as already remarked. For instance, with a small sample size $M \in \{27, 68, 163\}$, the lowest value of the mean KLD, achieved for $N \approx M/\log(M)$, remains in the order of $5 \cdot 10^{-2}$.

This behavior is evident also for $K \in \{4, 8, 16\}$, as illustrated in Fig. 5b, Fig. 6a, and Fig. 6b, respectively. Contrary to expectations, even with a small



(a) $K = 8$.



(b) $K = 16$.

Fig. 6: Accuracy of BPHs (with order N) in approximating Erlang PDFs with shape and rate equal to K: mean value of the KLD with respect to the ground truth, over 100 random samples of size M.

sample size $M \in \{27, 68\}$, for growing values of K we observe that accuracy improves as the Bernstein order N increases up to $N \in \{32, 64\}$, exceeding the $M/\log(M)$ threshold. This behavior may be attributed to the inherent complexity of the ground truth PDF, which increases with K, thus allowing BPHs to benefit from a higher order. Nevertheless, note that, with small sample sizes, the best achieved mean KLD remains significantly above the $5 \cdot 10^{-3}$ accuracy threshold. Overall, these results show how BPH approximants effectively support the selection of an ideal Bernstein order N based on the available sample size M, as well as a tradeoff between N and the evaluation accuracy.

5 Conclusions

We extended a consolidated method [3] for the selection of the order of BP approximants of CDFs with bounded support, exploiting it to select the order of BPH approximants of PDFs with unbounded support. We compared the accuracy of the approach with respect to Gaussian KDEs, considering a set of Erlang PDFs with expected value equal to 1. The results showed the effectiveness of the BPH approximant in balancing the bias-variance tradeoff, achieving better or comparable performance with respect to KDEs while maintaining a low memory footprint. In fact, the BPH closed-form representation only requires the storage of a number of parameters equal to the Bernstein order. Future work includes extending the experimental evaluation to other PDFs and optimizing the placement of the points used to construct the approximant BPH to cover the portion of the support where most of the probability mass is located.

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